

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 30****Topics covered :**

1. Introduction to Laplace Transform
2. Region of Convergence

- **Response of an linear time-invariant system to complex exponential:** The response of an linear time-invariant (LTI) system with impulse response $h(t)$ to a complex exponential $x(t) = e^{st}$ is given by

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \quad (1)$$

$$\begin{aligned} y(t) &= x(t) * h(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau \text{ (for an LTI system)} \\ &= \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}d\tau \\ &= e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau}_{\mathbf{H(s)}} \\ &= e^{st} \mathbf{H(s)} \end{aligned} \quad (2)$$

where $\mathbf{H(s)}$ is a complex constant, whose value depends on s . From the above derivation, it can be observed that the output is a constant times the input. Such functions are called [eigen functions](#) and the amplitude or constant factor is called the [eigen value](#).

- For the analysis of LTI systems, the usefulness of decomposing an input signal in-terms of eigen functions can be understood as follows. Consider an input $x(t)$ as

$$x(t) = \sum_i k_i e^{s_i t}$$

From the eigen function property, response of each of the individual inputs can be written as

$$k_i e^{s_i t} \rightarrow k_i \mathbf{H(s_i)} e^{s_i t}$$

If $\mathbf{H(s_i)}$ (which is not always guaranteed), the output $y(t)$ can be found using superposition and is given as

$$y(t) = \sum_i k_i \mathbf{H(s_i)} e^{s_i t}$$

- Determining the response of an LTI system to an input $x(t)$ which can be represented as a linear combination of complex exponential's e^{st} involves only two steps.

1. Compute the function $\mathbf{H}(\mathbf{s}) = \int_{-\infty}^{\infty} h(t)e^{-st} dt$.
2. Check if $\mathbf{H}(\mathbf{s})$ exists for all s_i and if so, compute the $\mathbf{H}(\mathbf{s})$
3. Compute the output using the linear combination of the responses.

- It can be seen that the output $y(t)$ can be expressed as a sum of complex exponential's e^{st} and the coefficient of each exponential is given by $\mathbf{H}(\mathbf{s}) = \int_{-\infty}^{\infty} h(t)e^{-st} dt$. Extending this idea to any signal, we define the Laplace transform of a signal $x(t)$ denoted by $\mathbf{X}(\mathbf{s})$ or $\mathcal{L}\{x(t)\}$ as

$$\mathbf{X}(\mathbf{s}) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-st} dt \quad (3)$$

where $\mathbf{X}(\mathbf{s})$ is a function of complex variable $\mathbf{s} = \sigma + j\omega$. This generalizes the concept of transforming a signal into a sum of complex exponentials, and the Laplace transform provides a powerful method for analyzing LTI systems.

- **Existence/Convergence of Laplace transform:** Laplace transformation can be viewed as the Fourier transformation of $x(t)e^{-\sigma t}$. Hence deriving from the conditions of convergence of Fourier transform, it can be stated that, the necessary condition for convergence of the Laplace transform is the absolute integrability of $x(t)e^{-\sigma t}$ i.e.

$$\int_{-\infty}^{\infty} |x(t)e^{-\sigma t}| dt < \infty \quad (4)$$

The range of σ for which the Laplace transform converges is called as [region of convergence](#) (ROC).

- **Laplace Transform of some common signals:**

- Impulse $\delta(t)$: The Laplace transform of an impulse signal is given by

$$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-st} dt = e^{-s0} = 1 \quad (5)$$

The ROC is the entire s -plane.

- Unit step $u(t)$: The Laplace transform of an unit step signal is given by

$$\mathcal{L}\{u(t)\} = \int_{-\infty}^{\infty} u(t)e^{-st} dt = \int_0^{\infty} e^{-st} dt = \frac{1}{s} \quad (6)$$

The Laplace transform exists if $\text{Re}\{s\} > 0$. Thus the ROC $\sigma > 0$.

- $e^{-at}u(t)$: The Laplace transform of an exponential decaying one-sided signal is given by

$$\mathcal{L}\{e^{-at}u(t)\} = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = \frac{1}{s+a} \quad (7)$$

The Laplace transform exists if $\text{Re}\{s+a\} > 0$. Thus the ROC $\sigma > -a$.